

SIMATS ENGINEERING

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Machine learning

Experiment 2

For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm in python to output a description of the set of all hypotheses consistent with the training examples

```
import pandas as pd
import numpy as np

# Read the uploaded CSV file
data = pd.read_csv("trainingdata (3).csv")

print("Training Data:\n")
print(data)

# Separate features and target
concepts = np.array(data.iloc[:, :-1])
target = np.array(data.iloc[:, -1])

# Candidate Elimination Algorithm
def candidate_elimination(concepts, target):

    specific_h = concepts[0].copy()

    general_h = [["?" for i in range(len(specific_h))]
                  for j in range(len(specific_h))]

    for i, h in enumerate(concepts):

        if target[i] == "Yes":
```

```

        if target[i] == "Yes":

            for x in range(len(specific_h)):
                if h[x] != specific_h[x]:
                    specific_h[x] = '?'
                    general_h[x][x] = '?'

        if target[i] == "No":

            for x in range(len(specific_h)):
                if h[x] != specific_h[x]:
                    general_h[x][x] = specific_h[x]
                else:
                    general_h[x][x] = '?'

        general_h = [g for g in general_h if g != ['?'] * len(specific_h)]

        return specific_h, general_h

s_final, g_final = candidate_elimination(concepts, target)

print("\nFinal Specific Hypothesis:")
print(s_final)

print("\nFinal General Hypothesis:")
for g in g_final:
    print(g)

```

OUTPUT :

```

... Training Data:

   Sky AirTemp Humidity   Wind Water Forecast EnjoySport
0  Sunny   Warm   Normal Strong  Warm   Same      Yes
1  Sunny   Warm    High Strong  Warm   Same      Yes
2  Rainy   Cold    High Strong  Warm Change     No
3  Sunny   Warm    High Strong  Cool   Same      Yes

Final Specific Hypothesis:
['Sunny' 'Warm' '?' 'Strong' '?' 'Same']

Final General Hypothesis:
['Sunny', '?', '?', '?', '?', '?']
['?', 'Warm', '?', '?', '?', '?']
['?', '?', '?', '?', '?', 'Same']

```

